34 Three statements about alpha (α) particles and beta (β) particles are listed.

1 The magnitude of the charge on an α-particle is twice the charge on a β-particle.

2 The mass of an α-particle is large compared with the mass of a β-particle.

3 The kinetic energy of an α-particle is small compared to the kinetic energy of a β-particle.

Which statements help to explain the different ionising effects of α-particles and β-particles?

A 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only